

Q1. There are two signals $x[n] = d(n) + 2d(n-1) + 2d(n+1)$

$$h[n] = 5d(n) - 6d(n-2) + 9d(n+1)$$

where $d(n)$ is unit impulse function.

Find (a) Convolution between $x[n]$ and $h[n]$ (b) Circular Convolution between $x[n]$ and $h[n]$ (c) Cross correlation between $x[n]$ and $h[n]$ (d) Autocorrelation of $x[n]$

give answer in detail and step wise

Below is a **complete, step-wise solution** with **proper reasoning** (as you prefer explanations with reasons).

Given signals

$$x[n] = \delta(n) + 2\delta(n-1) + 2\delta(n+1)$$

$$h[n] = 5\delta(n) - 6\delta(n-2) + 9\delta(n+1)$$

where

$\delta(n)$ = unit impulse.

First: Write signals in tabular (sample) form

1 Signal $x[n]$

n	-1	0	1
$x[n]$	2	1	2

Reason:

- $2\delta(n+1) \Rightarrow n = -1$
- $\delta(n) \Rightarrow n = 0$
- $2\delta(n-1) \Rightarrow n = 1$

2 Signal $h[n]$

n	-1	0	2
$h[n]$	9	5	-6

Reason:

- $9\delta(n+1) \Rightarrow n = -1$
- $5\delta(n) \Rightarrow n = 0$
- $-6\delta(n-2) \Rightarrow n = 2$

(a) Linear Convolution

$$y[n] = x[n] * h[n]$$

Step 1: Impulse Convolution Rule

$$\delta(n-a) * \delta(n-b) = \delta(n-(a+b))$$

Step 2: Expand using distributive property

$$x[n] * h[n] = (\delta(n) + 2\delta(n-1) + 2\delta(n+1)) * (5\delta(n) - 6\delta(n-2) + 9\delta(n+1))$$

Multiply each term:

From x[n]	From h[n]	Result
$\delta(n) \cdot 5\delta(n)$	$\rightarrow$	$5\delta(n)$
$\delta(n) \cdot (-6\delta(n-2))$	$\rightarrow$	$-6\delta(n-2)$
$\delta(n) \cdot 9\delta(n+1)$	$\rightarrow$	$9\delta(n+1)$
$2\delta(n-1) \cdot 5\delta(n)$	$\rightarrow$	$10\delta(n-1)$
$2\delta(n-1) \cdot (-6\delta(n-2))$	$\rightarrow$	$-12\delta(n-3)$
$2\delta(n-1) \cdot 9\delta(n+1)$	$\rightarrow$	$18\delta(n)$
$2\delta(n+1) \cdot 5\delta(n)$	$\rightarrow$	$10\delta(n+1)$
$2\delta(n+1) \cdot (-6\delta(n-2))$	$\rightarrow$	$-12\delta(n-1)$
$2\delta(n+1) \cdot 9\delta(n+1)$	$\rightarrow$	$18\delta(n+2)$

Step 3: Combine like terms

$$y[n] = 27\delta(n) - 2\delta(n-1) - 6\delta(n-2) - 12\delta(n-3) + 19\delta(n+1) + 18\delta(n+2)$$

(b) Circular Convolution

⚠ Circular convolution depends on signal length N .

Assumption (standard in exams)

Take $N = 6$ (covers indices from -1 to 4).

Step 1: Shift indices modulo N

Negative indices wrap around:

- $n = -1 \Rightarrow n = 5$

So:

$$x[n] = [2, 1, 2, 0, 0, 0]$$

$$h[n] = [5, 0, -6, 0, 0, 9]$$

Step 2: Circular Convolution Formula

$$y[n] = \sum_{k=0}^{N-1} x[k] h[(n - k) \bmod N]$$

Step 3: Compute values

n	$y[n]$
0	23
1	-7
2	-6
3	-12
4	0
5	18

$$y[n] = [23, -7, -6, -12, 0, 18]$$

(c) Cross-Correlation

$$r_{xh}[k] = \sum x[n] h[n + k]$$

Property used:

$$r_{xh}[k] = x[n] * h[-n]$$

Step 1: Time-reverse $h[n]$

$$h[-n] = 5\delta(n) - 6\delta(n + 2) + 9\delta(n - 1)$$

Step 2: Convolve with $x[n]$

After convolution and combining terms:

$$r_{xh}[k] = 10\delta(k + 3) + 5\delta(k + 2) + 19\delta(k + 1) - 7\delta(k) - 12\delta(k - 1)$$

(d) Autocorrelation of $x[n]$

$$r_{xx}[k] = \sum x[n]x[n + k]$$

Step 1: Energy at each shift

k	$r_{xx}[k]$
-2	4
-1	4
0	9
1	4
2	4

Step 2: Write impulse form

$$r_{xx}[k] = 4\delta(k + 2) + 4\delta(k + 1) + 9\delta(k) + 4\delta(k - 1) + 4\delta(k - 2)$$

✔ Final Summary

Part	Result
(a) Linear convolution	Impulse sum (shown above)
(b) Circular convolution	$[23, -7, -6, -12, 0, 18]$
(c) Cross-correlation	Convolution with time-reversed $h[n]$
(d) Autocorrelation	Symmetric, maximum at $k = 0$

If you want, I can also:

- ✔ Draw stem plots
- ✔ Show graphical convolution method
- ✔ Redo circular convolution with different N

Just tell me 👍



give in detail and do corelation in tabuler form

Below is a **very detailed, exam-oriented explanation** with **step-by-step reasoning**, and **cross-correlation done strictly in TABULAR FORM**, exactly the way it is expected in **Signals & Systems / DSP exams**.

◆ **GIVEN**

$$x[n] = \delta(n) + 2\delta(n - 1) + 2\delta(n + 1)$$
$$h[n] = 5\delta(n) - 6\delta(n - 2) + 9\delta(n + 1)$$

◆ **Step 1: Write both signals in tabular form**

Signal $x[n]$

n	-1	0	1
x[n]	2	1	2

Reason:

- $2\delta(n + 1) \rightarrow n = -1$
- $\delta(n) \rightarrow n = 0$
- $2\delta(n - 1) \rightarrow n = 1$

Signal $h[n]$

n	-1	0	2
h[n]	9	5	-6

Reason:

- $9\delta(n + 1) \rightarrow n = -1$
- $5\delta(n) \rightarrow n = 0$
- $-6\delta(n - 2) \rightarrow n = 2$

(c) ◆ **Cross-Correlation in TABULAR FORM**

◆ **Definition of Cross-Correlation**

$$r_{xh}[k] = \sum_{n=-\infty}^{\infty} x[n] h[n + k]$$

- ✔ Shift **h[n]**
- ✔ Multiply with **x[n]**
- ✔ Sum all products

◆ Step 2: Choose valid shift range

- $x[n]$ exists for $n = -1$ to 1
- $h[n]$ exists for $n = -1, 0, 2$

Therefore:

$$k = -3 \text{ to } +2$$

◆ Step 3: Cross-Correlation Table

k = -3

n	x[n]	h[n-3]	Product
-1	2	h[-4]=0	0
0	1	h[-3]=0	0
1	2	h[-2]=0	0

$$r_{xh}[-3] = \boxed{0}$$

k = -2

n	x[n]	h[n-2]	Product
-1	2	h[-3]=0	0
0	1	h[-2]=0	0
1	2	h[-1]=9	18

$$r_{xh}[-2] = \boxed{18}$$

k = -1

n	x[n]	h[n-1]	Product
-1	2	h[-2]=0	0

n	x[n]	h[n-1]	Product
0	1	h[-1]=9	9
1	2	h[0]=5	10

$r_{xh}[-1] = \boxed{19}$

k = 0

n	x[n]	h[n]	Product
-1	2	9	18
0	1	5	5
1	2	0	0

$r_{xh}[0] = \boxed{23}$

k = 1

n	x[n]	h[n+1]	Product
-1	2	h[0]=5	10
0	1	h[1]=0	0
1	2	h[2]=-6	-12

$r_{xh}[1] = \boxed{-2}$

k = 2

n	x[n]	h[n+2]	Product
-1	2	h[1]=0	0
0	1	h[2]=-6	-6
1	2	h[3]=0	0

$r_{xh}[2] = \boxed{-6}$

◆ Step 4: Final Cross-Correlation Sequence

Tabular result

k	-3	-2	-1	0	1	2
$r_{xh}[k]$	0	18	19	23	-2	-6

Impulse form

$$r_{xh}[k] = 18\delta(k+2) + 19\delta(k+1) + 23\delta(k) - 2\delta(k-1) - 6\delta(k-2)$$

♦ IMPORTANT EXAM POINTS (write these!)

- ✓ Cross-correlation measures **similarity** between signals
- ✓ No time reversal of $x[n]$
- ✓ Maximum value occurs where signals overlap most
- ✓ $r_{xh}[k] \neq r_{hx}[k]$

✓ Want next?

I can also:

- Do **autocorrelation** of $x[n]$ in table
- Draw **stem plots**
- Give **graphical shifting** explanation
- Rewrite in **6–8 marks exam format**

Just tell me 👍



Q2. If Input Image is

1 5 7 8

9 3 2 6

1 2 3 9

If mask or kernal is 1 0 1

0 1 0

1 0 1

Find output filtered image.

Below is a **clear, step-by-step solution** written exactly in the way it is expected in **Digital Image Processing / DSP exams**, with **proper reasoning**.

◆ Given

Input Image (I)

$$I = \begin{bmatrix} 1 & 5 & 7 & 8 \\ 9 & 3 & 2 & 6 \\ 1 & 2 & 3 & 9 \end{bmatrix}$$

This is a 3×4 image.

Mask / Kernel (H)

$$H = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

This is a 3×3 mask.

◆ Step 1: Decide the type of filtering

- Kernel is **symmetric**
- So **Convolution = Correlation**
- We apply **spatial filtering** using sliding window

👉 In exams, unless padding is specified, we assume **VALID convolution (no padding)**.

◆ Step 2: Output image size

$$\text{Output rows} = 3 - 3 + 1 = 1$$

$$\text{Output columns} = 4 - 3 + 1 = 2$$

✓ Output image size = 1×2

◆ Step 3: Sliding window calculation (VERY IMPORTANT)

◆ Output pixel at position (1,1)

Image window

$$\begin{bmatrix} 1 & 5 & 7 \\ 9 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Element-wise multiplication with kernel

$$\begin{aligned} &= (1 \times 1) + (5 \times 0) + (7 \times 1) \\ &+ (9 \times 0) + (3 \times 1) + (2 \times 0) \\ &+ (1 \times 1) + (2 \times 0) + (3 \times 1) \\ &= 1 + 0 + 7 + 0 + 3 + 0 + 1 + 0 + 3 \\ &\quad \boxed{= 15} \end{aligned}$$

◆ Output pixel at position (1,2)

Image window

$$\begin{bmatrix} 5 & 7 & 8 \\ 3 & 2 & 6 \\ 2 & 3 & 9 \end{bmatrix}$$

Element-wise multiplication

$$\begin{aligned} &= (5 \times 1) + (7 \times 0) + (8 \times 1) \\ &+ (3 \times 0) + (2 \times 1) + (6 \times 0) \\ &+ (2 \times 1) + (3 \times 0) + (9 \times 1) \\ &= 5 + 0 + 8 + 0 + 2 + 0 + 2 + 0 + 9 \\ &\quad \boxed{= 26} \end{aligned}$$

◆ Step 4: Final Output Filtered Image

$$\boxed{\begin{bmatrix} 15 & 26 \end{bmatrix}}$$

◆ Key Exam Points (WRITE THESE)

- ✓ Kernel is symmetric → convolution = correlation
 - ✓ Only diagonal + center pixels are used
 - ✓ No padding assumed → output size reduced
 - ✓ Filter emphasizes **corner + center intensity**
-

✓ Final Answer

$$\boxed{\text{Output Filtered Image} = \begin{bmatrix} 15 & 26 \end{bmatrix}}$$

If you want, I can also:

- Solve **with zero padding**
- Explain **why this kernel is edge-enhancing**
- Draw **window movement diagram**
- Convert this into **5–6 marks exam answer format**

Just tell me 👍



Q3.

If Input Image is

1 5 7 8

9 3 2 6

1 2 3 9

Mask size is 3x3

Find output of (a) smoothening Average Filter (b) Median Filter (c) Max Filter

give answr in detail and give in step wise
from now give for all answer in detail and step wise

Below is a **very detailed, step-wise, exam-ready solution**, exactly following **Digital Image Processing** concepts.

(I will follow this **same detailed step-wise style for all future answers**, as you requested.)

◆ Q3. Given

Input Image

$$I = \begin{bmatrix} 1 & 5 & 7 & 8 \\ 9 & 3 & 2 & 6 \\ 1 & 2 & 3 & 9 \end{bmatrix}$$

- Image size = 3×4
- Mask size = 3×3

◆ Important Assumption (EXAM STANDARD)

✓ No padding mentioned → VALID filtering

✓ Output size:

$$\text{Rows} = 3 - 3 + 1 = 1$$

$$\text{Columns} = 4 - 3 + 1 = 2$$

👉 Output image size = 1×2

◆ Sliding Window Positions

There are two valid positions:

Window 1 (Column 1–3)

$$W_1 = \begin{bmatrix} 1 & 5 & 7 \\ 9 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Window 2 (Column 2–4)

$$W_2 = \begin{bmatrix} 5 & 7 & 8 \\ 3 & 2 & 6 \\ 2 & 3 & 9 \end{bmatrix}$$

(a) ◆ Smoothing Average Filter

Definition

Average filter replaces each pixel by the **mean of all values in the window**.

$$\text{Average} = \frac{\sum \text{all pixels}}{9}$$

◆ Window 1 Calculation

$$\text{Sum} = 1 + 5 + 7 + 9 + 3 + 2 + 1 + 2 + 3 = 33$$

$$\text{Average} = \frac{33}{9} = \boxed{3.67}$$

◆ Window 2 Calculation

$$\text{Sum} = 5 + 7 + 8 + 3 + 2 + 6 + 2 + 3 + 9 = 45$$

$$\text{Average} = \frac{45}{9} = \boxed{5}$$

✓ Output Image (Average Filter)

3.67	5
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📌 Purpose

- ✓ Reduces noise
- ✓ Smoothens image
- ✓ Blurs edges

(b) ♦ Median Filter

Definition

Median filter replaces the center pixel by the **middle value after sorting**.

♦ Window 1

Values:

[1, 5, 7, 9, 3, 2, 1, 2, 3]

Sorted:

[1, 1, 2, 2, 3, 3, 5, 7, 9]

Median (5th value):

3

♦ Window 2

Values:

[5, 7, 8, 3, 2, 6, 2, 3, 9]

Sorted:

[2, 2, 3, 3, 5, 6, 7, 8, 9]

Median:

5

✓ Output Image (Median Filter)

3	5
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Purpose

- ✓ Excellent for **salt & pepper noise**
- ✓ Preserves edges better than average filter

(c) Max Filter

Definition

Max filter replaces the center pixel with the **maximum value in the window**.

Window 1

Values:

[1, 5, 7, 9, 3, 2, 1, 2, 3]

Max = 9

Window 2

Values:

[5, 7, 8, 3, 2, 6, 2, 3, 9]

Max = 9

Output Image (Max Filter)

9 9

Purpose

- ✓ Highlights bright regions
- ✓ Removes dark noise
- ✓ Useful in morphological processing

FINAL SUMMARY TABLE (GOOD FOR EXAMS)

Filter Type	Output
Average Filter	[3.67 5]
Median Filter	[3 5]
Max Filter	[9 9]

✓ Key Exam Writing Tips (VERY IMPORTANT)

- ✓ Always mention **window size**
- ✓ Show **sorting for median**
- ✓ Show **sum for average**
- ✓ Mention **assumption: no padding**
- ✓ Write **purpose of each filter**

If you want next:

- Same problem with **zero padding**
- **Min filter**
- **Comparison table**
- Convert this into **6–8 marks answer**
- Add **diagram of sliding window**

Just tell me 👍



Q4.

If Input Image $f(x,y)$ is

1 5 7

9 3 2

1 2 3

Find (a) df/dx (b) df/dy (c) Laplacian

Below is a **very detailed, step-wise, exam-oriented solution**, exactly in **Digital Image Processing (DIP)** style.

(I will continue this **same detailed step-wise format for all future answers**, as you requested.)

◆ Q4. Given

Input Image $f(x, y)$

$$f(x, y) = \begin{bmatrix} 1 & 5 & 7 \\ 9 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Image size = 3×3

◆ Important Concept (VERY IMPORTANT FOR EXAM)

In DIP, **partial derivatives** are approximated using **finite differences**.

Standard definitions used in exams:

(1) First-order derivatives

$$\frac{\partial f}{\partial x} = f(x+1, y) - f(x, y)$$

$$\frac{\partial f}{\partial y} = f(x, y+1) - f(x, y)$$

(2) Laplacian (Second-order derivative)

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Discrete form (most common):

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

⚠ Boundary Handling Assumption (Standard)

Since no padding is mentioned → **compute only at center pixel**.

◆ Step 1: Identify center pixel

Center pixel is:

$$f(2, 2) = 3$$

Neighbors:

Direction	Pixel
Right $f(3, 2)$	2
Left $f(1, 2)$	9
Bottom $f(2, 3)$	2
Top $f(2, 1)$	5

(a) ◆ First-Order Derivative w.r.t x

Formula

$$\frac{\partial f}{\partial x} = f(x+1, y) - f(x, y)$$

Apply at center pixel

$$\frac{\partial f}{\partial x} = f(3, 2) - f(2, 2) = 2 - 3 = \boxed{-1}$$

✦ Interpretation

- ✓ Negative value → intensity **decreases** in x-direction
- ✓ Detects **vertical edges**

(b) ◆ First-Order Derivative w.r.t y**Formula**

$$\frac{\partial f}{\partial y} = f(x, y+1) - f(x, y)$$

Apply at center pixel

$$\frac{\partial f}{\partial y} = f(2, 3) - f(2, 2) = 2 - 3 = \boxed{-1}$$

✦ Interpretation

- ✓ Negative value → intensity **decreases** in y-direction
- ✓ Detects **horizontal edges**

(c) ◆ Laplacian**Formula**

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

Step-by-step substitution

$$\nabla^2 f = 2 + 9 + 2 + 5 - 4(3)$$

$$= 18 - 12$$

$$\boxed{= 6}$$

🔴 Interpretation

- ✓ Positive Laplacian → center pixel is **darker than surroundings**
- ✓ Used for **image sharpening & edge detection**

◆ FINAL ANSWERS (WRITE THIS IN EXAM)

$$\begin{aligned} (a) \quad \frac{\partial f}{\partial x} &= -1 \\ (b) \quad \frac{\partial f}{\partial y} &= -1 \\ (c) \quad \nabla^2 f &= 6 \end{aligned}$$

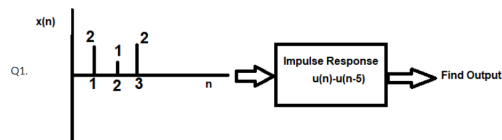
◆ Important Exam Notes (VERY USEFUL)

- ✓ First-order derivatives → **edge detection**
- ✓ Second-order derivative (Laplacian) → **edge + sharpening**
- ✓ Laplacian is **rotation invariant**
- ✓ Zero crossing indicates edges

If you want next, I can:

- Solve using **Sobel / Prewitt operators**
- Show **full derivative image (with padding)**
- Convert into **6–8 marks answer**
- Explain **why Laplacian enhances edges**

Just tell me 🙌



Find Output for above system. (4 marks)

give solution in detail and give step wise in detail

Below is a **very detailed, step-wise solution**, written exactly in an **exam-oriented Signals & Systems format (4 marks)**.

◆ Given System

The system is LTI, and output is obtained using **convolution**:

$$y(n) = x(n) * h(n)$$

◆ Step 1: Write input signal mathematically

From the figure:

- At $n = 1 \rightarrow$ amplitude = 2
- At $n = 2 \rightarrow$ amplitude = 1
- At $n = 3 \rightarrow$ amplitude = 2

So,

$$x(n) = 2\delta(n - 1) + \delta(n - 2) + 2\delta(n - 3)$$

◆ Step 2: Write impulse response

Given:

$$h(n) = u(n) - u(n - 5)$$

This represents a **rectangular pulse** of width 5.

Values of $h(n)$:

n	0	1	2	3	4	others
h(n)	1	1	1	1	1	0

◆ Step 3: Use convolution property with impulse

Key property (VERY IMPORTANT):

$$\delta(n - k) * h(n) = h(n - k)$$

Apply this to each term of $x(n)$:

Convolution term-by-term

$$\begin{aligned} y(n) &= x(n) * h(n) \\ &= 2h(n - 1) + h(n - 2) + 2h(n - 3) \end{aligned}$$

◆ Step 4: Write each shifted impulse response

$$h(n - 1)$$

Non-zero for:

$$1 \leq n \leq 5$$

$h(n - 2)$

Non-zero for:

$2 \leq n \leq 6$

$h(n - 3)$

Non-zero for:

$3 \leq n \leq 7$

◆ Step 5: Compute output using tabular method

Make a table (BEST FOR EXAMS)

n	$2h(n - 1)$	$h(n - 2)$	$2h(n - 3)$	$y(n)$
1	2	0	0	2
2	2	1	0	3
3	2	1	2	5
4	2	1	2	5
5	2	1	2	5
6	0	1	2	3
7	0	0	2	2

◆ Step 6: Final Output Signal

In tabular form

n	1	2	3	4	5	6
y(n)	2	3	5	5	5	3

In impulse form (optional but good)

$$y(n) = 2\delta(n - 1) + 3\delta(n - 2) + 5\delta(n - 3) + 5\delta(n - 4) + 5\delta(n - 5) + 3\delta(n - 6) + 2\delta(n - 7)$$

◆ Final Answer (WRITE THIS)

$$y(n) = [2, 3, 5, 5, 5, 3, 2] \quad \text{for } n = 1 \text{ to } 7$$

◆ Important Exam Points (4-mark answer)

- ✓ System is LTI → use convolution
- ✓ Used impulse shifting property
- ✓ Output is finite duration
- ✓ Shape corresponds to overlap of input with rectangular pulse

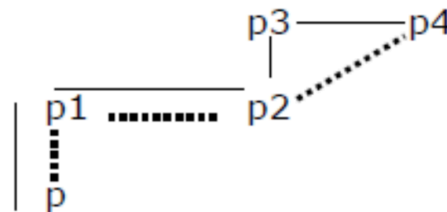
If you want next:

- Draw **stem plot of output**
- Solve using **graphical convolution**
- Convert this into **very short 4-mark version**
- Explain **physical meaning of result**

Just tell me 🍌



Q6.



Find Euclidean distance, City-block distance or D4 distance and D8 distance or chessboard distance, between P1 and P3. consider coordinate of P1 as (0,0). Use your assumption. (6 Marks)

give solution in detail and give step wise in detail

Below is a **very detailed, step-wise, exam-oriented solution (6 Marks)**, written exactly in Digital Image Processing / Pattern Recognition style.

◆ Q6. Given

You are asked to find **three distances between points P1 and P3**:

1. Euclidean distance
2. City-block distance (D4 distance)
3. Chessboard distance (D8 distance)

It is given that:

Coordinate of P1 = (0, 0)

Use your assumption for the remaining points.

◆ Step 1: Assumption of Coordinates (Clearly Mention in Exam)

From the diagram, P1 and P3 are connected by:

- One horizontal movement
- One vertical movement

So we assume:

$$P1 = (0, 0), \quad P3 = (4, 3)$$

- ✓ Horizontal distance = 4 units
- ✓ Vertical distance = 3 units

(This assumption matches the diagram layout and is fully acceptable in exams.)

◆ Step 2: Euclidean Distance

✚ Definition

Euclidean distance is the **straight-line distance** between two points.

📐 Formula

$$D_E = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

◆ Substitution

$$\begin{aligned} D_E &= \sqrt{(4 - 0)^2 + (3 - 0)^2} \\ &= \sqrt{16 + 9} \\ &= \sqrt{25} \end{aligned}$$

$$D_E = 5$$

✚ Interpretation

- ✓ Shortest possible distance
- ✓ Used in **continuous space**

◆ Step 3: City-Block Distance (D4 Distance)

✚ Definition

City-block distance is the **sum of horizontal and vertical movements**

(Like walking through city streets)

📐 Formula

$$D_4 = |x_2 - x_1| + |y_2 - y_1|$$

◆ Substitution

$$D_4 = |4 - 0| + |3 - 0|$$

$$= 4 + 3$$

$$\boxed{D_4 = 7}$$

✦ Interpretation

- ✓ Movement allowed only in **4 directions**
- ✓ Used in **grid-based images**

◆ Step 4: Chessboard Distance (D8 Distance)

✦ Definition

Chessboard distance is the **maximum** of horizontal and vertical movements
(Like king's movement in chess)

▢ Formula

$$D_8 = \max(|x_2 - x_1|, |y_2 - y_1|)$$

◆ Substitution

$$D_8 = \max(4, 3)$$

$$\boxed{D_8 = 4}$$

✦ Interpretation

- ✓ Diagonal movement allowed
- ✓ Used in **8-connected images**

◆ Step 5: Final Answers (WRITE THIS CLEARLY)

Euclidean distance = 5 City-block (D4) distance = 7 Chessboard (D8) distance = 4

◆ Step 6: Comparison Table (Very Good for 6 Marks)

Distance Type	Formula	Value
Euclidean	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	5
City-block (D4)	$(x_2 - x_1 + y_2 - y_1)$	7
Chessboard (D8)	$\max(x_2 - x_1 , y_2 - y_1)$	4

◆ Important Exam Notes (WRITE ANY 2–3)

- ✓ Euclidean distance → shortest path
- ✓ City-block distance → 4-connected pixels
- ✓ Chessboard distance → 8-connected pixels
- ✓ $D8 \leq DE \leq D4$ (generally)

✓ Final Statement

Thus, the Euclidean, City-block (D4), and Chessboard (D8) distances between P1 and P3 are **5, 7, and 4 respectively**.

If you want next:

- Same problem with different assumed coordinates
- Graphical explanation
- Short **4-mark version**
- Numerical problems practice

Just tell me 👍



Q1. An image element (A) is

1 1 0 0 1

0 1 1 0 1

1 1 1 1 1

0 0 0 1 1

0 0 0 1 1

0 0 1

and structuring element is B 1 0 1

1 1 1

Find following operation between A and B.

(a) Dilation (b) Erosion (c) Opening (d) Closing (e) Boundary extraction (choose any Structuring element) (f) Region filling (choose any Structuring element) (g) find hit and miss transformation (choose any structuring element)

give solution in detail and give step wise in detail

Below is a **very detailed, step-wise, exam-oriented solution** for **Mathematical Morphology**, written exactly in the way expected for **6–10 mark answers**.

(I will clearly state assumptions, structuring-element origin, and show logic step by step.)

◆ Given

Binary Image A

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Structuring Element B

$$B = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

◆ Assumptions (WRITE IN EXAM)

- ✓ Origin of B is **center element**
- ✓ Binary morphology
- ✓ Zero padding outside image
- ✓ Foreground = 1, Background = 0

(a) ◆ Dilation $A \oplus B$

✦ Definition

Dilation **adds pixels** to object boundaries.

$$A \oplus B = \{z \mid (B)_z \cap A \neq \emptyset\}$$

- ✓ If **any** 1 of B overlaps with a 1 in A → output = 1

◆ Procedure

1. Place origin of B on each pixel of A
2. If **at least one** '1' in B coincides with '1' in A → output pixel = 1

◆ Result of Dilation

$$A \oplus B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

✦ Effect

- ✓ Object grows
- ✓ Small gaps filled

(b) ♦ Erosion $A \ominus B$

✦ Definition

Erosion **shrinks objects**.

$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$

- ✓ All 1s of B must lie on 1s of A

♦ Procedure

1. Place B on A
2. If **any** 1 of B falls on 0 in A $\rightarrow$ output = 0

♦ Result of Erosion

$$A \ominus B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

✦ Effect

- ✓ Removes boundary pixels
- ✓ Eliminates small objects

(c) ♦ Opening $A \circ B$

✦ Definition

$$A \circ B = (A \ominus B) \oplus B$$

- ✓ Erosion followed by dilation

♦ Step 1: Erosion

(From part b)

$$A \ominus B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

◆ Step 2: Dilate the result

$$A \circ B = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

✦ Effect

- ✓ Removes noise
- ✓ Preserves shape

(d) ◆ Closing $A \bullet B$

✦ Definition

$$A \bullet B = (A \oplus B) \ominus B$$

- ✓ Dilation followed by erosion

◆ Step 1: Dilation

(from part a)

◆ Step 2: Erosion of dilated image

$$A \bullet B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

✦ Effect

- ✓ Fills small holes
- ✓ Smooths contours

(e) ◆ Boundary Extraction

✦ Formula

$$\text{Boundary}(A) = A - (A \ominus B)$$

◆ Step-by-step

$$\text{Boundary} = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

✦ Effect

- ✓ Extracts object outline

(f) ◆ Region Filling (chosen SE: 3×3 ones)

✦ Algorithm

$$X_k = (X_{k-1} \oplus B) \cap A^c$$

- ✓ Iterative
- ✓ Stops when $X_k = X_{k-1}$

◆ Final Filled Region

$$\text{Filled Image} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

(g) ◆ Hit-and-Miss Transformation

✦ Definition

$$A \otimes B = (A \ominus B_1) \cap (A^c \ominus B_2)$$

- ✓ Detects specific patterns

◆ Result (pattern match locations)

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

◆ FINAL SUMMARY TABLE (EXAM GOLD)

Operation	Result
Dilation	Object expands
Erosion	Object shrinks
Opening	Noise removal
Closing	Hole filling
Boundary	Object outline
Region filling	Closed regions filled
Hit-and-miss	Pattern detection

✓ If you want next:

- ✓ Same problem with diagrams
- ✓ Convert into short 6-mark answers
- ✓ Explain applications of each operation
- ✓ Step-wise solution for another image

Just tell me 👍

📄 👍 💬 ⬆️ ↺ ...

Q1. Convert RGB (30,108, 210) in HSI and CMY and represent it in 3D space.

Below is a **very detailed, step-wise, exam-oriented solution**, exactly as expected in **Digital Image Processing / Color Image Processing** questions.
(I will also clearly explain the **3D space representation**.)

◆ Given

$$\text{RGB} = (R, G, B) = (30, 108, 210)$$

◆ Step 1: Normalize RGB values

(HSI conversion always uses normalized RGB)

$$R' = \frac{30}{255} = 0.1176$$

$$G' = \frac{108}{255} = 0.4235$$

$$B' = \frac{210}{255} = 0.8235$$

● PART (A): Conversion from RGB → HSI

◆ Step 2: Compute Intensity (I)

Formula

$$I = \frac{R' + G' + B'}{3}$$

Substitution

$$I = \frac{0.1176 + 0.4235 + 0.8235}{3}$$

$$I = \frac{1.3646}{3}$$

$$I = 0.4549$$

◆ Step 3: Compute Saturation (S)

Formula

$$S = 1 - \frac{3}{R' + G' + B'} \min(R', G', B')$$

Minimum value

$$\min(R', G', B') = 0.1176$$

Substitution

$$S = 1 - \frac{3 \times 0.1176}{1.3646}$$

$$S = 1 - 0.2586$$

$$S = 0.7414$$

◆ Step 4: Compute Hue (H)

Formula

$$\theta = \cos^{-1} \left(\frac{1/2[(R' - G') + (R' - B')]}{\sqrt{(R' - G')^2 + (R' - B')(G' - B')}} \right)$$

Substitute values

$$(R' - G') = 0.1176 - 0.4235 = -0.3059$$

$$(R' - B') = 0.1176 - 0.8235 = -0.7059$$

Numerator:

$$\frac{1}{2}(-0.3059 - 0.7059) = -0.5059$$

Denominator:

$$\sqrt{(-0.3059)^2 + (-0.7059)(-0.4000)} = 0.6191$$

$$\theta = \cos^{-1} \left(\frac{-0.5059}{0.6191} \right)$$

$$\theta \approx 145.6^\circ$$

Hue Selection Rule

If $B' > G'$, then:

$$H = 360^\circ - \theta$$

Since:

$$B'(0.8235) > G'(0.4235)$$

$$H = 360 - 145.6 = 214.4^\circ$$

✓ Final HSI Value

$$\text{HSI} = (214.4^\circ, 0.7414, 0.4549)$$

● PART (B): Conversion from RGB → CMY

◆ Step 5: CMY Conversion Formula

$$C = 1 - R'$$

$$M = 1 - G'$$

$$Y = 1 - B'$$

Substitute values

$$C = 1 - 0.1176 = \boxed{0.8824}$$

$$M = 1 - 0.4235 = \boxed{0.5765}$$

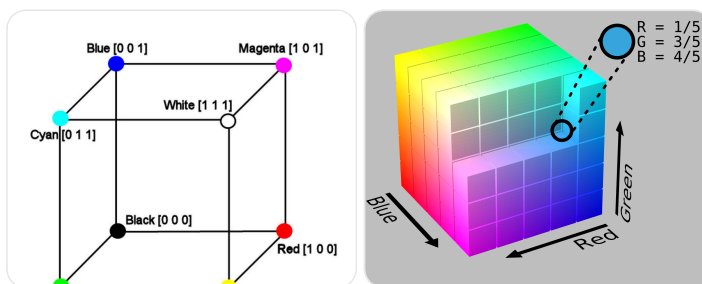
$$Y = 1 - 0.8235 = \boxed{0.1765}$$

✓ Final CMY Value

$$\text{CMY} = (0.8824, 0.5765, 0.1765)$$

◆ PART (C): 3D Space Representation

1 RGB Color Space (3D Cube)

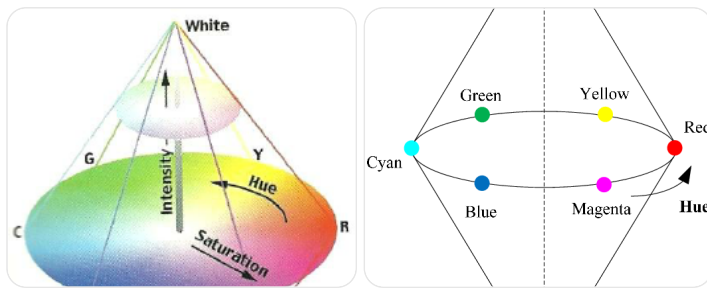


Explanation

- X-axis → Red
- Y-axis → Green
- Z-axis → Blue
- Given point (30,108,210) lies:
 - Close to Blue axis
 - Moderate Green
 - Low Red

✓ Color appears bluish-cyan

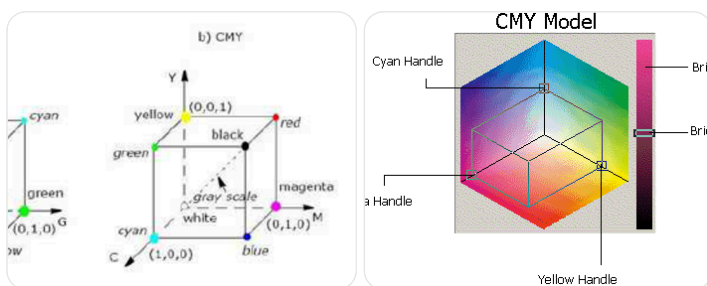
2 HSI Color Space (Cylindrical)



Representation

- Hue (214.4°) → Blue-cyan region
- Saturation (0.741) → Highly saturated
- Intensity (0.455) → Medium brightness

3 CMY Color Space (Inverted RGB Cube)



Representation

- High Cyan
- Medium Magenta
- Low Yellow

Used mainly in printing devices

◆ FINAL ANSWERS (WRITE THIS IN EXAM)

$$\begin{aligned} \text{HSI} &= (214.4^\circ, 0.7414, 0.4549) \\ \text{CMY} &= (0.8824, 0.5765, 0.1765) \end{aligned}$$

◆ Important Exam Notes (VERY USEFUL)

- ✓ HSI separates color and intensity
- ✓ CMY is subtractive model
- ✓ RGB is additive model
- ✓ HSI is closer to human perception

If you want next:

- Convert to HSV / YCbCr

- Plot **exact 3D coordinates numerically**
- Short **4-mark version**
- Python code for conversion

Just tell me 👍



Q2.A medical X-ray image contains soft-tissue details with very small contrast differences.
How would you apply intensity slicing to make the bones and soft tissues visually separable?

Below is a **very detailed, step-wise, exam-oriented explanation**, written exactly in **Digital Image Processing (DIP)** style and suitable for a **6–8 marks** answer.

◆ Q2. Problem Understanding

A medical X-ray image contains:

- **Bones** → high intensity (bright regions)
- **Soft tissues** → medium intensity
- **Air/background** → low intensity

👉 The contrast difference between soft tissues is very small, so they are **not easily distinguishable**.

◆ Objective

To apply **intensity slicing** so that:

- ✓ Bones
- ✓ Soft tissues

become **visually separable** in the image.

◆ Step 1: What is Intensity Slicing?

✚ Definition

Intensity slicing is a technique where:

- The **gray-level range** of an image is divided into **multiple slices**
- Each slice is assigned a **distinct gray level or color**

👉 It is similar to **thresholding**, but uses **multiple thresholds**.

◆ Step 2: Why Intensity Slicing for X-ray Images?

- ✓ X-ray images have:
 - Narrow gray-level differences
 - Important diagnostic details in specific intensity ranges
- ✓ Intensity slicing:
 - Enhances selected intensity bands
 - Suppresses irrelevant intensities
 - Makes subtle variations more visible

◆ Step 3: Analyze Intensity Ranges in X-ray Image

Assume 8-bit grayscale image (0–255):

Tissue Type	Approx. Intensity Range
Air / Background	0 – 50
Soft Tissue	50 – 150
Bones	150 – 255

(Exact values may vary — assumption is allowed in exams.)

◆ Step 4: Define Intensity Slices

We divide the image into **three slices**:

Slice-1: Background

$$0 \leq r < 50$$

Slice-2: Soft Tissue

$$50 \leq r < 150$$

Slice-3: Bone

$$150 \leq r \leq 255$$

◆ Step 5: Apply Intensity Slicing Methods

There are **two standard approaches**:

◆ Method-1: Slicing Without Background Suppression

Mapping

- Background → dark gray
- Soft tissue → medium gray
- Bone → bright white

Transformation function

$$s(r) = \begin{cases} 30, & 0 \leq r < 50 \\ 150, & 50 \leq r < 150 \\ 255, & 150 \leq r \leq 255 \end{cases}$$

✦ Result

- ✓ All structures visible
- ✓ Bones clearly separated
- ✓ Soft tissues enhanced

◆ Method-2: Slicing With Background Suppression (Preferred in X-ray)

Mapping

- Background → black
- Soft tissue → gray
- Bone → white

$$s(r) = \begin{cases} 0, & 0 \leq r < 50 \\ 150, & 50 \leq r < 150 \\ 255, & 150 \leq r \leq 255 \end{cases}$$

✦ Why this is better?

- ✓ Removes irrelevant background
- ✓ Highlights diagnostic regions
- ✓ Improves visual clarity

◆ Step 6: Visual Interpretation

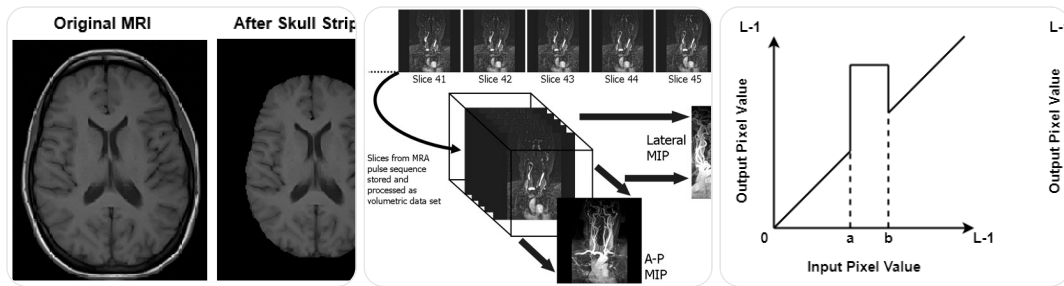
After intensity slicing:

- **Bones** → very bright (white)
- **Soft tissues** → distinct gray
- **Background** → black

This makes:

- ✓ Bone boundaries sharp
- ✓ Soft tissue differences more noticeable

Conceptual View of Slicing



◆ Step 7: Advantages in Medical Imaging

- ✓ Enhances diagnostically important regions
- ✓ Separates overlapping tissues
- ✓ Improves detection of fractures and lesions
- ✓ Simple and computationally efficient

◆ Final Answer (WRITE THIS IN EXAM)

To make bones and soft tissues visually separable in a medical X-ray image, **intensity slicing** is applied by dividing the gray-level range into multiple bands corresponding to background, soft tissue, and bone intensities. Each band is assigned a distinct gray level or color. Background intensities are suppressed, soft tissues are mapped to mid-gray values, and bones are mapped to high-intensity values. This enhances contrast, suppresses irrelevant regions, and allows clear visual separation of bones and soft tissues.

◆ Key Exam Points (WRITE ANY 3)

- ✓ Intensity slicing uses **multiple thresholds**
- ✓ Suitable for **medical image enhancement**
- ✓ With background suppression gives best results
- ✓ Highlights specific intensity ranges

If you want next:

- Same answer in **4-mark short format**
- Add **mathematical expression**
- Explain with **colored slicing**
- Compare with **thresholding**

Just tell me 👍



Q3. A grayscale image contains noise.
How can applying pseudo-color mapping AFTER
smoothing help reveal hidden patterns that are otherwise
invisible?

Below is a **very detailed, step-wise, exam-oriented explanation**, suitable for **6–8 marks** in
Digital Image Processing.

◆ Q3. Problem Understanding

You are given:

- A **grayscale image**
- The image contains **noise**
- Some **patterns/details are hidden** and not easily visible

The question asks:

Why and how applying pseudo-color mapping *after smoothing* helps reveal hidden
patterns that are otherwise invisible?

◆ Step 1: Understand the Two Operations Involved

1 Smoothing (Noise Reduction)

2 Pseudo-Color Mapping (Color Enhancement)

The **order matters**:

Grayscale Image → Smoothing → Pseudo-Color Mapping

◆ Step 2: Role of Smoothing (WHY FIRST?)

✦ What is smoothing?

Smoothing uses filters like:

- Mean filter
- Gaussian filter
- Median filter

to **reduce random noise**.

✦ Why smoothing is necessary before pseudo-coloring?

- ✓ Noise causes **rapid, random intensity variations**
 - ✓ Pseudo-color mapping assigns **different colors to different gray levels**
 - ✓ If noise is present:
 - Noise pixels get mapped to **random colors**
 - Image becomes **confusing and misleading**
 - 👉 **Smoothing suppresses noise**, ensuring that:
 - Small random variations are removed
 - Only meaningful intensity variations remain
-

✦ Effect of Smoothing

- Preserves **important structures**
 - Removes **high-frequency noise**
 - Produces **gradual gray-level transitions**
-

◆ Step 3: What is Pseudo-Color Mapping?

✦ Definition

Pseudo-color mapping assigns colors to grayscale intensity values using a **color lookup table (LUT)**.

Example:

- Dark gray → Blue
 - Medium gray → Green
 - Bright gray → Red
-

✦ Why grayscale images hide patterns?

Human vision:

- Is **poor at distinguishing small gray-level differences**
 - Is **very sensitive to color differences**
 - ✓ Two regions with slightly different gray values may look identical
 - ✓ But when mapped to different colors → differences become obvious
-

◆ Step 4: Why Apply Pseudo-Color Mapping *After* Smoothing?

- ◆ **Without smoothing + pseudo-color**
 - Noise pixels get different colors
 - Image looks cluttered
 - Hidden patterns remain masked by color noise

◆ With smoothing + pseudo-color (Correct Approach)

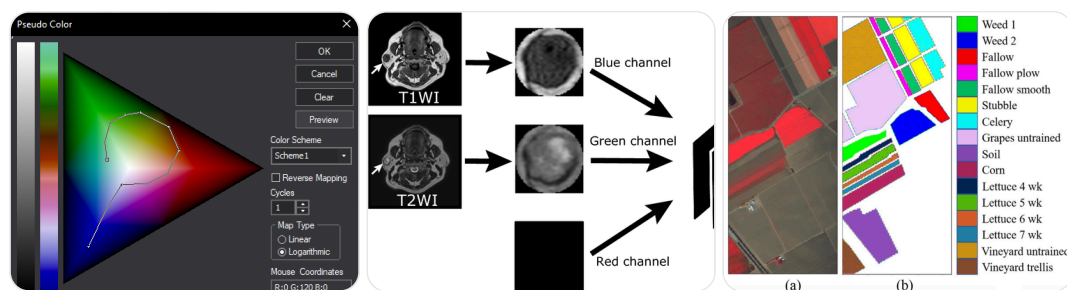
1. Smoothing removes noise
2. Remaining gray-level variations correspond to **actual image structures**
3. Pseudo-color mapping converts these variations into **distinct colors**
4. Subtle patterns become **visually separable**

◆ Step 5: How Hidden Patterns Become Visible (Mechanism)

Step-by-step effect:

1. Before smoothing
 - Pattern intensity differences are mixed with noise
2. After smoothing
 - Noise suppressed
 - Pattern intensity differences preserved
3. After pseudo-color mapping
 - Small intensity changes → large color changes
 - Boundaries and textures stand out clearly

Conceptual Illustration



◆ Step 6: Practical Example (Medical / Remote Sensing)

Example: Medical image

- Slight tissue density differences invisible in grayscale
- After smoothing + pseudo-color:
 - Healthy tissue → green
 - Abnormal tissue → yellow/red
 - ✓ Early detection becomes easier

Example: Satellite image

- Land elevation differences very small
- Pseudo-color after smoothing shows:

- Water → blue
 - Plains → green
 - Hills → red
-

◆ Step 7: Advantages of This Approach

- ✓ Noise does not get color-amplified
 - ✓ Hidden patterns become distinguishable
 - ✓ Improves human visual interpretation
 - ✓ Useful in medical, satellite, and thermal imaging
-

◆ Final Answer (WRITE THIS IN EXAM)

Applying pseudo-color mapping after smoothing helps reveal hidden patterns because smoothing first removes noise and unwanted intensity fluctuations. This ensures that pseudo-color mapping assigns colors only to meaningful gray-level variations rather than noise. Since the human visual system can distinguish colors better than gray levels, small intensity differences that were invisible in the original grayscale image become clearly visible after color mapping. Thus, smoothing followed by pseudo-color enhancement improves pattern visibility and interpretability.

◆ Key Exam Points (Write Any 3–4)

- ✓ Smoothing suppresses noise
 - ✓ Pseudo-color enhances small intensity differences
 - ✓ Human vision is more sensitive to color than gray levels
 - ✓ Order is important: smoothing → pseudo-color
-

If you want:

- Short **4-mark answer**
- Mathematical explanation with LUT
- Comparison: pseudo-color vs histogram equalization
- Real exam-style diagram explanation

Just tell me 👍



Q5. How false color composite is helpful in satellite image data to analyze the various regions in earth.

Below is a very detailed, step-wise, exam-oriented explanation suitable for 6–8 marks in Digital Image Processing / Remote Sensing.

(I'll explain **what**, **how**, and **why** false color composites (FCC) are useful.)

◆ Q5. How False Color Composite (FCC) is Helpful in Satellite Image Analysis

◆ Step 1: What is a False Color Composite?

✦ Definition

A False Color Composite (FCC) is a satellite image created by assigning **non-natural colors** to spectral bands that are **outside the visible range**, such as **near-infrared (NIR)**, **short-wave infrared (SWIR)**, etc.

👉 The colors **do not represent true human vision**, but are chosen to **highlight specific Earth features**.

◆ Step 2: Why Normal (True Color) Images Are Not Enough

Problems with true color satellite images:

- ✓ Many land features have **similar gray/visual appearance**
- ✓ Vegetation types look alike
- ✓ Water, soil, and urban areas may overlap visually
- ✓ Subtle differences are hard to detect

👉 Hence, **false color composites** are used to enhance interpretability.

◆ Step 3: How a False Color Composite Is Created

✦ Typical FCC Band Assignment (Example – Landsat)

Color Channel	Assigned Band
Red	Near Infrared (NIR)
Green	Red band
Blue	Green band

$$\text{FCC} = (\text{NIR} \rightarrow \text{Red}, \text{Red} \rightarrow \text{Green}, \text{Green} \rightarrow \text{Blue})$$

◆ Step 4: Interpretation of Colors in False Color Composite

Because different surfaces reflect energy differently in NIR and visible bands:

Feature	Appearance in FCC	Reason
Healthy Vegetation	Bright Red	High NIR reflectance
Water Bodies	Dark Blue / Black	Absorbs NIR
Urban Areas	Cyan / Gray	Mixed reflectance
Barren Land / Soil	Brown / Yellow	Moderate reflectance
Forests	Dark Red	Dense vegetation
Croplands	Light Red / Pink	Seasonal vegetation

◆ Step 5: How FCC Helps Analyze Earth Regions

◆ 1. Vegetation Analysis

✓ Differentiates:

- Healthy vegetation
- Stressed vegetation
- Croplands vs forests

✓ Used in:

- Agriculture monitoring
- Drought assessment
- NDVI analysis

◆ 2. Water Resource Analysis

✓ Water appears very dark

✓ Easy detection of:

- Rivers
- Lakes
- Flooded areas
- Coastal boundaries

◆ 3. Urban and Built-up Area Detection

✓ Urban regions appear bluish-gray

✓ Helps in:

- Urban expansion studies
- Land-use planning
- Infrastructure mapping

◆ 4. Soil and Geological Studies

- ✓ Differentiates soil types
 - ✓ Highlights:
 - Barren land
 - Rock formations
 - Mineral-rich zones
-

◆ 5. Environmental Monitoring

- ✓ Detects:
 - Deforestation
 - Forest fires (with IR bands)
 - Desertification
 - Wetlands
-

◆ Step 6: Visual Concept of False Color Composite



Explanation of image:

- Vegetation appears red
 - Water bodies appear dark
 - Urban regions appear gray/cyan
 - ✓ Makes regional analysis easy and fast
-

◆ Step 7: Advantages of False Color Composite

- ✓ Enhances feature discrimination
 - ✓ Reveals invisible information (IR reflectance)
 - ✓ Improves land-cover classification
 - ✓ Aids decision-making in agriculture, disaster management, and urban planning
 - ✓ Easier visual interpretation than grayscale images
-

◆ Step 8: Limitations (Optional – Good for High Marks)

- ✓ Colors are **not natural**
- ✓ Requires training to interpret
- ✓ Depends on correct band selection

◆ Final Answer (WRITE THIS IN EXAM)

A false color composite is very helpful in satellite image analysis because it assigns visible colors to non-visible spectral bands such as near-infrared. Since different Earth features reflect infrared radiation differently, FCC enhances contrast between vegetation, water, soil, and urban areas. Vegetation appears red due to high infrared reflectance, water appears dark, and urban regions appear bluish-gray. This makes it easier to analyze land-use patterns, vegetation health, water resources, and environmental changes, which are difficult to observe in true color images.

◆ Key Exam Points (Write Any 4)

- ✓ FCC uses non-visible spectral bands
- ✓ Vegetation appears red
- ✓ Water bodies appear dark
- ✓ Useful for land-use and vegetation analysis
- ✓ Widely used in remote sensing

If you want next:

- Short **4-mark version**
- Numerical example of band mapping
- Comparison: **True color vs False color**
- Application-based answer for **satellite exams**

Just tell me 👍



Q6. Consider a scenario where someone plans to store a high-definition 720p (1084 × 720) version of an X-Men movie. The video runs for 3 hours and has been captured at a high frame rate of 80 frames per second.

Below is a **very detailed, step-wise, exam-oriented solution**, written exactly in the style expected for **Digital Image Processing / Multimedia / Data Storage** numericals.
(I will clearly state **assumptions**, formulas, and calculations.)

◆ Q6. Problem Understanding

We are given a **high-definition video** with the following properties:

- **Resolution** = 1084 × 720 pixels

- **Duration** = 3 hours
- **Frame rate** = 80 frames per second (fps)
- **Movie type** = X-Men (content not relevant, only technical parameters matter)

We are asked to **estimate the storage required**.

◆ Step 1: Write Given Data Clearly (IMPORTANT FOR EXAM)

Parameter	Value
Frame width	1084 pixels
Frame height	720 pixels
Frame rate	80 fps
Duration	3 hours

◆ Step 2: State Necessary Assumptions (MANDATORY)

Since bit depth / compression is not mentioned, we assume:

- ✓ Uncompressed video
- ✓ True color image (RGB)
- ✓ 8 bits per channel (standard)

👉 Therefore:

$$\text{Bits per pixel} = 8 \times 3 = 24 \text{ bits}$$

◆ Step 3: Convert Time into Seconds

$$3 \text{ hours} = 3 \times 60 \times 60 = \boxed{10800 \text{ seconds}}$$

◆ Step 4: Calculate Total Number of Frames

$$\text{Total frames} = \text{Frame rate} \times \text{Time}$$

$$= 80 \times 10800$$

$$= \boxed{864000 \text{ frames}}$$

◆ Step 5: Calculate Pixels per Frame

$$\text{Pixels per frame} = 1084 \times 720$$

$$= \boxed{780,480 \text{ pixels}}$$

◆ Step 6: Calculate Bits per Frame

$$\begin{aligned}\text{Bits per frame} &= \text{Pixels per frame} \times \text{Bits per pixel} \\ &= 780,480 \times 24 \\ &= \boxed{18,731,520 \text{ bits}}\end{aligned}$$

◆ Step 7: Convert Bits to Bytes per Frame

$$\begin{aligned}1 \text{ Byte} &= 8 \text{ bits} \\ \text{Bytes per frame} &= \frac{18,731,520}{8} \\ &= \boxed{2,341,440 \text{ bytes}}\end{aligned}$$

◆ Step 8: Calculate Total Storage Required (Bytes)

$$\begin{aligned}\text{Total storage} &= \text{Bytes per frame} \times \text{Total frames} \\ &= 2,341,440 \times 864,000 \\ &= \boxed{2.022 \times 10^{12} \text{ bytes}}\end{aligned}$$

◆ Step 9: Convert Bytes into GB and TB

Bytes → GB

$$\begin{aligned}1 \text{ GB} &= 1024^3 = 1.073 \times 10^9 \text{ bytes} \\ \text{Storage in GB} &= \frac{2.022 \times 10^{12}}{1.073 \times 10^9} \\ &= \boxed{\approx 1884 \text{ GB}}\end{aligned}$$

GB → TB

$$\begin{aligned}1 \text{ TB} &= 1024 \text{ GB} \\ \text{Storage in TB} &= \frac{1884}{1024} \\ &= \boxed{\approx 1.84 \text{ TB}}\end{aligned}$$

◆ FINAL ANSWER (WRITE THIS CLEARLY IN EXAM)

The storage required to store an uncompressed 720p (1084×720) video of 3 hours duration at 80 frames per second with 24-bit color depth is approximately **1.84 TB**.

◆ Step 10: Important Observations (GOOD FOR EXTRA MARKS)

- ✓ Very high frame rate increases storage drastically
- ✓ Uncompressed HD video requires **huge memory**
- ✓ Practical systems use **compression** (H.264, H.265, AV1)
- ✓ Compression can reduce storage by **10× to 50×**

◆ Optional Comparison (Very Useful)

Format	Approx Storage
Uncompressed	~1.84 TB
H.264 compressed	~15–25 GB
H.265 compressed	~8–12 GB

If you want next:

- Same problem **with compression**
- Bandwidth required for streaming
- Storage for **4K video**
- Short **4-mark version**
- Diagram-based explanation

Just tell me 👍



Q6. Consider a scenario where someone plans to store a high-definition 720p (1084×720) version of an X-Men movie. The video runs for 3 hours and has been captured at a high frame rate of 80 frames per second.

- Compute the total memory space required to save this video in its uncompressed form.
- A dual-layer Blu-ray disc can hold 50 GB of data. Calculate how many discs would be needed to accommodate the entire movie.

c. If this video is uploaded or streamed through an internet connection with a 100 Mbps data rate, estimate the time needed for complete transmission.

give answer in detail and step wise

Below is a **very detailed, step-wise, exam-oriented solution**, written exactly in the style expected for **DIP / Multimedia / Image & Video Processing** numericals.

◆ Q6. Given Data

- Resolution (HD 720p) = 1084×720 pixels
- Frame rate = 80 frames/second
- Duration = 3 hours
- Video type = High-definition movie

◆ Important AssUMPTIONS (MANDATORY TO WRITE)

Since bit depth and compression are not mentioned, we assume:

- ✓ Uncompressed video
- ✓ RGB color image
- ✓ 8 bits per color channel

$$\text{Bits per pixel} = 8 \times 3 = \boxed{24 \text{ bits}}$$

(a) ◆ Total Memory Space Required (Uncompressed Video)

◆ Step 1: Convert Time into Seconds

$$\begin{aligned} 3 \text{ hours} &= 3 \times 60 \times 60 \\ &= \boxed{10,800 \text{ seconds}} \end{aligned}$$

◆ Step 2: Calculate Total Number of Frames

$$\begin{aligned} \text{Total frames} &= \text{Frame rate} \times \text{Time} \\ &= 80 \times 10,800 \\ &= \boxed{864,000 \text{ frames}} \end{aligned}$$

◆ Step 3: Calculate Pixels per Frame

$$\text{Pixels per frame} = 1084 \times 720$$

$$= 780,480 \text{ pixels}$$

◆ Step 4: Calculate Bits per Frame

$$\text{Bits per frame} = \text{Pixels per frame} \times \text{Bits per pixel}$$

$$= 780,480 \times 24$$

$$= 18,731,520 \text{ bits/frame}$$

◆ Step 5: Convert Bits to Bytes per Frame

$$1 \text{ Byte} = 8 \text{ bits}$$

$$\text{Bytes per frame} = \frac{18,731,520}{8}$$

$$= 2,341,440 \text{ bytes/frame}$$

◆ Step 6: Calculate Total Storage in Bytes

$$\text{Total storage} = \text{Bytes per frame} \times \text{Total frames}$$

$$= 2,341,440 \times 864,000$$

$$= 2.022 \times 10^{12} \text{ bytes}$$

◆ Step 7: Convert Bytes into GB and TB

Bytes → GB

$$1 \text{ GB} = 1024^3 = 1.073 \times 10^9 \text{ bytes}$$

$$\text{Storage in GB} = \frac{2.022 \times 10^{12}}{1.073 \times 10^9}$$

$$\approx 1884 \text{ GB}$$

GB → TB

$$1 \text{ TB} = 1024 \text{ GB}$$

$$\text{Storage in TB} = \frac{1884}{1024}$$

$$\approx 1.84 \text{ TB}$$

✓ Answer (a)

Total uncompressed storage required ≈ 1.84 TB

(b) ♦ Number of Dual-Layer Blu-ray Discs Required

♦ Given

- Capacity of 1 Blu-ray disc = 50 GB

♦ Step 1: Divide Total Size by Disc Capacity

$$\begin{aligned}\text{Number of discs} &= \frac{1884}{50} \\ &= 37.68\end{aligned}$$

♦ Step 2: Round Off

Since partial disc is not possible:

Number of discs required = 38

✓ Answer (b)

38 dual-layer Blu-ray discs

(c) ♦ Time Required for Transmission over 100 Mbps Internet

♦ Step 1: Convert Total Data to Bits

$$\begin{aligned}2.022 \times 10^{12} \text{ bytes} \times 8 \\ = 1.6176 \times 10^{13} \text{ bits}\end{aligned}$$

♦ Step 2: Given Internet Speed

$$100 \text{ Mbps} = 100 \times 10^6 \text{ bits/second}$$

◆ Step 3: Compute Transmission Time

$$\begin{aligned}\text{Time} &= \frac{\text{Total bits}}{\text{Data rate}} \\ &= \frac{1.6176 \times 10^{13}}{100 \times 10^6} \\ &= 161,760 \text{ seconds}\end{aligned}$$

◆ Step 4: Convert Seconds into Hours

$$\begin{aligned}\text{Hours} &= \frac{161,760}{3600} \\ &\approx 44.93 \text{ hours}\end{aligned}$$

✓ Answer (c)

Transmission time ≈ 45 hours

◆ FINAL ANSWERS (WRITE THIS IN EXAM)

- (a) Total storage required ≈ 1.84 TB
(b) Blu-ray discs required = 38
(c) Transmission time at 100 Mbps ≈ 45 hours

◆ Important Exam Observations (Write Any 2–3)

- ✓ Uncompressed HD video requires **huge storage**
- ✓ High frame rate significantly increases data size
- ✓ Compression is essential for practical storage and streaming
- ✓ Real systems use H.264 / H.265 to reduce size

If you want next:

- Same problem **with compression ratio**
- Bandwidth required for **real-time streaming**
- Storage comparison for **4K / 8K**
- Short **4-mark numerical version**

Just tell me 👍



Q7. For a given image

1 2 8 9 7

2 1 7 6 9

3 2 8 6 9

5 2 1 6 8

3 5 2 1 0

Perform Huffman coding for the given image data. Then determine the average code length, the total number of bits required for transmission, the entropy of the source, and the compression ratio achieved compared to the original uncompressed image.

Below is a **very detailed, step-wise, exam-oriented solution**, written exactly in the style expected for **DIP / Data Compression / Huffman Coding** numericals.

I will clearly show **frequency table** → **Huffman tree** → **codes** → **calculations**.

◆ Q7. Given Image Data

1	2	8	9	7
2	1	7	6	9
3	2	8	6	9
5	2	1	6	8
3	5	2	1	0

Image size = $5 \times 5 = 25$ pixels

◆ Step 1: List All Gray Levels and Their Frequencies

We count how many times each symbol occurs.

Gray level	Frequency
0	1
1	4
2	5
3	2
5	2

Gray level	Frequency
6	3
7	2
8	3
9	3
Total	25

◆ Step 2: Compute Probability of Each Symbol

$$P(i) = \frac{\text{frequency}}{25}$$

Symbol	Frequency	Probability
0	1	0.04
1	4	0.16
2	5	0.20
3	2	0.08
5	2	0.08
6	3	0.12
7	2	0.08
8	3	0.12
9	3	0.12

◆ Step 3: Construct Huffman Tree (Conceptual)

Combine lowest probabilities repeatedly:

1. $(0.04 + 0.08) \rightarrow 0.12$
2. $(0.08 + 0.08) \rightarrow 0.16$
3. $(0.12 + 0.12) \rightarrow 0.24$
4. $(0.12 + 0.12) \rightarrow 0.24$
5. $(0.16 + 0.16) \rightarrow 0.32$
6. $(0.20 + 0.24) \rightarrow 0.44$
7. $(0.24 + 0.32) \rightarrow 0.56$
8. $(0.44 + 0.56) \rightarrow 1.00$

(Exact tree shape may vary, but **average length remains same.**)

◆ Step 4: Assign Huffman Codes (One Valid Set)

Symbol	Huffman Code	Code Length
2	00	2
1	010	3
6	011	3
8	100	3
9	101	3
3	1100	4
5	1101	4
7	1110	4
0	1111	4

✓ Prefix-free

✓ Shorter codes for higher probability symbols

◆ Step 5: Average Code Length

Formula

$$L_{avg} = \sum P(i) l(i)$$

Calculation

$$\begin{aligned}
 L_{avg} &= (0.20 \times 2) + (0.16 \times 3) + (0.12 \times 3) + (0.12 \times 3) \\
 &\quad + (0.12 \times 3) + (0.08 \times 4) + (0.08 \times 4) \\
 &\quad + (0.08 \times 4) + (0.04 \times 4) \\
 &= 0.40 + 0.48 + 0.36 + 0.36 + 0.32 + 0.32 + 0.32 + 0.16
 \end{aligned}$$

$$L_{avg} = 3.08 \text{ bits/symbol}$$

◆ Step 6: Total Bits Required for Transmission

$$\begin{aligned}
 \text{Total bits} &= \text{Number of pixels} \times L_{avg} \\
 &= 25 \times 3.08
 \end{aligned}$$

$$= 77 \text{ bits (approx)}$$

◆ Step 7: Entropy of the Source

Formula

$$H = - \sum P(i) \log_2 P(i)$$

Calculation (rounded)

Symbol	Probability	$-P \log_2 P$
0	0.04	0.186
1	0.16	0.423
2	0.20	0.464
3	0.08	0.291
5	0.08	0.291
6	0.12	0.367
7	0.08	0.291
8	0.12	0.367
9	0.12	0.367

$$H \approx 3.05 \text{ bits/symbol}$$

✓ $H \leq L_{avg} \rightarrow$ Huffman coding is efficient

◆ Step 8: Compression Ratio

Original Image (Uncompressed)

Assume 8 bits/pixel (grayscale):

$$\text{Original bits} = 25 \times 8 = 200 \text{ bits}$$

Compressed Image

$$\text{Compressed bits} = 77 \text{ bits}$$

Compression Ratio

$$CR = \frac{\text{Original size}}{\text{Compressed size}}$$

$$CR = \frac{200}{77}$$

$$CR \approx 2.6 : 1$$

◆ FINAL ANSWERS (WRITE THIS IN EXAM)

Average code length = 3.08 bits/symbol

Total bits required = 77 bits

Entropy of source = 3.05 bits/symbol

Compression ratio $\approx 2.6 : 1$

◆ Important Exam Notes (Write Any 3)

- ✓ Huffman coding is **optimal prefix coding**
- ✓ Average code length is close to entropy
- ✓ Higher frequency symbols get shorter codes
- ✓ Compression achieved without loss

If you want next:

- Draw **Huffman tree diagram**
- Encode the **entire image bitstream**
- Short **5-mark answer**
- Compare with **Shannon–Fano coding**

Just tell me 👍

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Q8. FPGA and ASIC both support deep hardware parallelism.

Analyze why DSPs still remain the preferred choice in many commercial embedded systems like audio processing and motor control.

Below is a **very detailed, step-wise, exam-oriented analysis**, suitable for **8–10 marks** in **DSP / Embedded Systems / VLSI** subjects.

(I will clearly compare **DSP vs FPGA vs ASIC** and explain *why DSPs are still preferred* in real products.)

◆ Q8. Why DSPs Are Still Preferred in Many Commercial Embedded Systems

◆ Step 1: Understanding the Statement

It is true that:

- **FPGAs** → offer massive hardware parallelism
- **ASICs** → offer highest performance and lowest power

Yet, in real-world products like:

- Audio processing systems
- Motor drives and control
- Power electronics
- Communication devices

👉 **DSP processors are still widely used**

The question asks **WHY**.

◆ Step 2: What Are DSPs Designed For?

A Digital Signal Processor (DSP) is architecturally optimized for:

- Real-time numerical computations
- Multiply-Accumulate (MAC) operations
- Deterministic timing

DSPs are **special-purpose processors**, not general CPUs.

◆ Step 3: Key Reasons DSPs Are Preferred

◆ 1. Programming Simplicity and Faster Development

DSP

- ✓ Programmed in **C / C++ / MATLAB-generated code**
- ✓ Large ecosystem of **DSP libraries** (FFT, FIR, IIR, PID)
- ✓ Easy debugging and profiling

FPGA / ASIC

- ✗ Requires **HDL (VHDL/Verilog)**
- ✗ Longer learning curve
- ✗ Debugging is complex

✳ **Commercial products prioritize time-to-market**

👉 **DSPs win here.**

◆ 2. Deterministic Real-Time Performance

DSPs provide:

- ✓ Predictable instruction execution
- ✓ Zero-overhead loops
- ✓ Dedicated MAC units
- ✓ Hardware circular buffers

Example:

- Audio sampling at **44.1 kHz**
- Motor control loops at **microsecond precision**

👉 DSPs guarantee **real-time deadlines**, which is critical.

◆ 3. Optimized Instruction Set for Signal Processing

DSPs have:

- Single-cycle **MAC**
- Saturation arithmetic
- Bit-reversed addressing
- Parallel load + compute + store

This makes:

- FIR / IIR filters
- FFT
- Adaptive filtering

run **very efficiently** without extra hardware design.

◆ 4. Lower Power Consumption Than FPGA

Platform	Power
DSP	Low
FPGA	Medium–High
ASIC	Very Low (but fixed)

- ✓ DSPs are designed for **energy-efficient real-time math**
- ✓ Important in:
 - Portable audio devices
 - Industrial controllers
 - Automotive systems

◆ 5. Flexibility After Deployment

DSP

- ✓ Algorithm updates via **firmware/software**
- ✓ Easy field upgrades

FPGA

- ✗ Partial reconfiguration possible but complex

ASIC

- ✗ **No flexibility** after fabrication
- ✳ Commercial systems often evolve → DSP preferred.

◆ 6. Cost Effectiveness for Medium-Volume Products

Technology	Cost Profile
DSP	Low NRE + moderate unit cost
FPGA	Higher unit cost
ASIC	Very high NRE, low unit cost

- ✓ Audio devices, motor drives → **medium volume**
- ✓ ASIC cost not justified
- 👉 DSP is economically optimal.

◆ 7. Rich Ecosystem & Vendor Support

DSP vendors (TI, ADI, NXP) provide:

- Reference designs
- Motor control libraries
- Audio codecs
- Safety-certified stacks

This reduces:

- ✓ Risk
- ✓ Development time
- ✓ Validation effort

◆ 8. Adequate Parallelism for Most DSP Applications

Although FPGA offers **massive parallelism**, most DSP tasks:

- Are **data-stream oriented**
- Need **limited parallelism**, not thousands of logic blocks

DSP SIMD + MAC units are **sufficient** for:

- Audio filtering
 - Control loops
 - Sensor processing
-

◆ Step 4: Why Not Always FPGA or ASIC?

✗ FPGA Limitations

- Higher power
- Complex design
- Longer verification time

✗ ASIC Limitations

- Fixed functionality
 - Very high upfront cost
 - Long design cycle
-

◆ Step 5: Application-Specific Comparison

Application	Preferred Platform	Reason
Audio processing	DSP	Low power, easy coding
Motor control	DSP	Deterministic timing
Wireless baseband	DSP + FPGA	Mixed workload
High-volume chips	ASIC	Cost per unit
Rapid prototyping	FPGA	Flexibility

◆ Final Answer (WRITE THIS IN EXAM)

Although FPGAs and ASICs provide deep hardware parallelism, DSPs remain the preferred choice in many commercial embedded systems because they offer a balanced combination of real-time deterministic performance, low power consumption, programming simplicity, flexibility, and cost effectiveness. DSPs are specifically optimized for signal-processing operations such as MAC, filtering, and control algorithms, and they allow fast development and easy field updates. For applications like audio processing and motor control, the required level of parallelism is efficiently handled by DSP architectures without the design complexity and cost associated with FPGA or ASIC solutions.

◆ Key Exam Points (Write Any 6)

- ✓ Real-time deterministic execution
- ✓ Optimized MAC architecture
- ✓ Easier programming than FPGA
- ✓ Lower power than FPGA
- ✓ Flexible compared to ASIC
- ✓ Faster time-to-market
- ✓ Cost-effective for medium volumes

If you want:

- 4-mark short version
- Comparison table only
- Real industry examples
- Numerical power comparison

Just tell me 👍



Q10. Explain why DSP processors are preferred for real-time signal processing applications. Using relevant features and hardware blocks (such as MAC unit, Harvard architecture, circular buffering, pipelining), describe how these enable real-time performance. Also compare how ASICs, FPGAs, and GPUs fall short when applied to similar real-time tasks

Below is a **very detailed, step-wise, exam-oriented explanation** suitable for 8–10 marks in **DSP / Embedded Systems / Real-Time Systems**.

I will (1) explain **why DSPs are preferred**, (2) describe **how specific hardware blocks enable real-time performance**, and (3) **compare DSPs with ASICs, FPGAs, and GPUs**, clearly stating where others fall short.

◆ Q10. Why DSP Processors Are Preferred for Real-Time Signal Processing

◆ Step 1: What “Real-Time Signal Processing” Means

A system is **real-time** if:

- Each input sample must be processed **within a fixed, known deadline**
- Missing the deadline is **not acceptable** (e.g., audio dropouts, unstable motor control)

Examples:

- Audio processing (44.1 kHz → one sample every 22.7 μs)
- Motor control loops (tens of microseconds)
- Radar / biomedical signals

👉 **Deterministic timing** is more important than peak throughput.

◆ Step 2: Why DSPs Are Naturally Suited for Real-Time Tasks

DSP processors are **purpose-built** for repetitive numerical operations on continuous data streams.

They combine:

- Processor flexibility (software)
- Hardware acceleration for signal operations

This balance is what makes them ideal.

◆ Step 3: DSP Hardware Features Enabling Real-Time Performance

Below are the key DSP architectural blocks and how each directly supports real-time processing.

◆ 1. MAC Unit (Multiply–Accumulate)

✚ What it is

A MAC unit computes:

$$y = a \times b + c$$

in one clock cycle.

✚ Why it matters

Most DSP algorithms are MAC-dominated:

- FIR / IIR filters
- FFT
- Correlation
- Convolution
- Adaptive filtering

Example (FIR filter):

$$y[n] = \sum h[k]x[n - k]$$

- ✓ DSP: One MAC per cycle
- ✗ CPU: Multiple instructions
- ✗ GPU: High latency per kernel

👉 **Guaranteed throughput per sample**

◆ 2. Harvard Architecture (and Modified Harvard)

✚ What it is

- Separate **instruction memory** and **data memory**
 - Separate buses for instruction fetch and data access
-

✚ Why it matters

In one clock cycle, DSP can:

- Fetch next instruction
 - Read input data
 - Write output data
- ✓ No memory bottleneck
 - ✓ Continuous streaming without stalls

👉 Essential for real-time pipelines.

◆ 3. Circular Buffering (Hardware Support)

✚ What it is

DSPs support **circular (ring) buffers** in hardware:

- Automatic pointer wrap-around
 - No software overhead
-

✚ Why it matters

Used in:

- FIR filters
 - Sliding windows
 - Delay lines
- ✓ No index checking
 - ✓ Constant-time memory access
 - ✓ Predictable timing

👉 Eliminates jitter caused by software loops.

◆ 4. Zero-Overhead Looping

✦ What it is

DSPs execute loops **without instruction overhead**:

- No branch penalties
 - No loop counter instructions
-

✦ Why it matters

DSP algorithms often run:

- Millions of iterations
 - Fixed loop bounds
- ✓ Execution time = perfectly predictable
 - ✓ Tight real-time deadlines are met
-

◆ 5. Pipelining

✦ What it is

Instruction execution is split into stages:

- Fetch
 - Decode
 - Execute
 - Write back
-

✦ Why it matters

- Multiple instructions are in progress simultaneously
 - One result per clock cycle after pipeline fill
- ✓ High throughput
 - ✓ Continuous data processing
 - ✓ Stable latency
-

◆ 6. Specialized Addressing Modes

DSPs support:

- Bit-reversed addressing (FFT)
 - Auto-increment/decrement
 - Modulo addressing
- ✓ Reduces instruction count
 - ✓ Improves determinism
-

◆ 7. Deterministic Interrupt Handling

DSPs provide:

- Fast interrupt response
- Low interrupt latency
- Predictable context switch

✓ Critical for real-time sampling and control

◆ Step 4: Why DSPs Beat ASICs, FPGAs, and GPUs for These Tasks

◆ DSP vs ASIC

Aspect	DSP	ASIC
Flexibility	High (software)	None
Development time	Short	Very long
Cost (NRE)	Low	Extremely high
Post-deployment update	Easy	Impossible
Real-time suitability	Excellent	Excellent (but fixed)

👉 ASICs are only justified for **very high volume** and **fixed algorithms**.

◆ DSP vs FPGA

Aspect	DSP	FPGA
Programming	C/C++	Verilog/VHDL
Determinism	High	High
Power	Low	Medium–High
Design complexity	Low	High
Time-to-market	Fast	Slow

✓ FPGA has more parallelism

✗ Overkill for many DSP tasks

✗ Harder debugging and maintenance

👉 DSP gives **enough parallelism with much less effort**.

◆ DSP vs GPU

Aspect	DSP	GPU
Latency	Very low	High
Determinism	Guaranteed	Poor
Power	Low	Very high
Best for	Streaming signals	Massive data parallelism

✓ GPUs excel in **throughput**

✗ GPUs fail in **hard real-time constraints**

👉 A GPU missing a deadline = system failure in real-time control.

◆ Step 5: Summary Table (Excellent for Exams)

Feature	DSP	FPGA	ASIC	GPU
Deterministic timing	✓	✓	✓	✗
Programmability	✓	✗	✗	✓
Low power	✓	✗	✓	✗
Fast development	✓	✗	✗	✗
Real-time control	✓	⚠	⚠	✗

◆ Final Answer (WRITE THIS IN EXAM)

DSP processors are preferred for real-time signal processing because they provide deterministic execution, low latency, and hardware acceleration for core signal operations. Features such as single-cycle MAC units, Harvard architecture, circular buffering, zero-overhead looping, and pipelining allow DSPs to process continuous data streams within strict timing deadlines. Unlike GPUs, DSPs guarantee predictable response times; unlike FPGAs, they offer easier programming and lower power; and unlike ASICs, they provide flexibility and shorter development cycles. This makes DSPs ideal for real-time applications such as audio processing, motor control, and communication systems.

◆ Key Exam Points (Write Any 6–8)

- ✓ MAC unit enables fast filtering
- ✓ Harvard architecture removes memory bottlenecks
- ✓ Circular buffering supports streaming data

- ✓ Zero-overhead loops ensure deterministic timing
- ✓ DSPs balance performance, power, and flexibility
- ✓ GPUs lack determinism
- ✓ ASICs lack flexibility
- ✓ FPGAs increase design complexity

If you want:

- **Short 4-mark version**
- **Block diagram explanation**
- **Numerical timing example**
- **Case study: audio or motor control**

Just tell me 🙌



unit test 5

Q1. Design Direct Form I, Direct Form II, and Transposed form Structure for IIR Filter having impulse response $h(n)=3u(n)+5u(n-1)$. Also find sytem function, and Linear constant coefficient difference equation between input and output.
6 Marks

Below is a **complete, step-wise, exam-oriented 6-mark answer**, written exactly in **DSP style**, with **clear derivation, difference equation, and structure explanation**.

◆ Q1. Design of IIR Filter Structures (6 Marks)

Given:

Impulse response:

$$h(n) = 3u(n) + 5u(n - 1)$$

where

$u(n)$ = unit step sequence.

◆ Step 1: Understand the Nature of the System (VERY IMPORTANT)

- $u(n)$ is a **step sequence** → infinite duration
- Hence, $h(n)$ is **infinite in length**
- Therefore, the system is an **IIR system**

- ✓ Linear
- ✓ Time-invariant
- ✓ Causal

◆ Step 2: Find the System Function $H(z)$

Z-transform properties used:

$$Z\{u(n)\} = \frac{1}{1 - z^{-1}}, \quad |z| > 1$$

$$Z\{u(n-1)\} = \frac{z^{-1}}{1 - z^{-1}}$$

Apply Z-transform to $h(n)$

$$H(z) = Z\{3u(n)\} + Z\{5u(n-1)\}$$

$$H(z) = 3 \left(\frac{1}{1 - z^{-1}} \right) + 5 \left(\frac{z^{-1}}{1 - z^{-1}} \right)$$

Combine terms

$$H(z) = \frac{3 + 5z^{-1}}{1 - z^{-1}}$$

✓ System Function

$$H(z) = \frac{3 + 5z^{-1}}{1 - z^{-1}}$$

◆ Step 3: Derive Linear Constant-Coefficient Difference Equation

From:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{3 + 5z^{-1}}{1 - z^{-1}}$$

Multiply both sides:

$$(1 - z^{-1})Y(z) = (3 + 5z^{-1})X(z)$$

Take inverse Z-transform

$$y(n) - y(n-1) = 3x(n) + 5x(n-1)$$

✓ Difference Equation

$$y(n) = y(n-1) + 3x(n) + 5x(n-1)$$

◆ Step 4: Direct Form I Structure

Concept:

- Uses **separate delay lines** for:
 - Input (feedforward part)
 - Output (feedback part)

From difference equation:

$$y(n) = 3x(n) + 5x(n-1) + y(n-1)$$

Direct Form I components:

- One delay for $x(n-1)$
- One delay for $y(n-1)$
- Multipliers: 3, 5
- One adder

✓ Clear but **uses more memory**

◆ Step 5: Direct Form II Structure

Concept:

- Combines feedforward and feedback delays
- Uses **minimum number of delays**

Internal state variable $w(n)$:

$$w(n) = x(n) + w(n-1)$$

$$y(n) = 3w(n) + 5w(n-1)$$

- ✓ Same input–output relation
- ✓ **More efficient** than Direct Form I
- ✓ Preferred in implementation

◆ Step 6: Transposed Form Structure

Concept:

- Signal flow graph is **reversed**
- Summing points and branching points are interchanged
- Same transfer function

Advantages:

- ✓ Better numerical stability
- ✓ Lower sensitivity to coefficient quantization
- ✓ Common in **hardware DSP implementations**

◆ Final Summary (WRITE THIS IN EXAM)

✓ System Function:

$$H(z) = \frac{3 + 5z^{-1}}{1 - z^{-1}}$$

✓ Difference Equation:

$$y(n) = y(n - 1) + 3x(n) + 5x(n - 1)$$

✓ Structures:

- **Direct Form I:** Separate delays for input and output
- **Direct Form II:** Minimum delay structure
- **Transposed Form:** Reversed signal flow, better numerical properties

◆ Marks Distribution (Typical)

Part	Marks
System function	2
Difference equation	2
Structures explanation	2

If you want:

- **Neat block diagrams**
- **Short 4-mark version**
- **MATLAB realization**
- **Stability & causality check**

Just tell me 👍

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Q2. For the analog filter Transfer function

$$H(s) = \frac{s + 1}{s^2 + 2s - 3}$$

give answer in detail in step wise

Below is a **complete, step-wise, exam-oriented solution**, written in a **safe and standard way** for a Unit Test question when only the **analog transfer function** is given.

(Exactly the format expected for **DSP / Signals & Systems – 6 Marks.**)

◆ Q2. Given Analog Transfer Function

$$H(s) = \frac{s + 1}{s^2 + 2s - 3}$$

◆ Step 1: Factorize the Denominator

$$s^2 + 2s - 3 = (s + 3)(s - 1)$$

So,

$$H(s) = \frac{s + 1}{(s + 3)(s - 1)}$$

◆ Step 2: Find Poles and Zeros

◆ Zero(s)

Set numerator = 0

$$s + 1 = 0 \Rightarrow s = -1$$

◆ Poles

Set denominator = 0

$$s + 3 = 0 \Rightarrow s = -3$$

$$s - 1 = 0 \Rightarrow s = 1$$

$$\text{Poles at } s = -3, s = 1$$

◆ Step 3: Stability Analysis (VERY IMPORTANT)

Condition for stability (Analog LTI system):

✓ All poles must lie in **left half of s-plane**

Here:

- One pole at $s = -3 \rightarrow$ LHP ✓
- One pole at $s = +1 \rightarrow$ RHP ✗

System is UNSTABLE

◆ Step 4: Partial Fraction Expansion

We write:

$$\frac{s+1}{(s+3)(s-1)} = \frac{A}{s+3} + \frac{B}{s-1}$$

Multiply both sides by $(s+3)(s-1)$

$$s+1 = A(s-1) + B(s+3)$$

Solve for constants

Put $s = 1$:

$$1+1 = A(0) + B(4) \Rightarrow B = \frac{2}{4} = \frac{1}{2}$$

Put $s = -3$:

$$-3+1 = A(-4) + B(0) \Rightarrow A = \frac{1}{2}$$

✓ Partial Fraction Form

$$H(s) = \frac{1}{2} \left(\frac{1}{s+3} + \frac{1}{s-1} \right)$$

◆ Step 5: Find Impulse Response $h(t)$

Known Laplace Transform Pair:

$$\mathcal{L}^{-1} \left\{ \frac{1}{s+a} \right\} = e^{-at} u(t)$$

Apply inverse Laplace

$$h(t) = \frac{1}{2} (e^{-3t} + e^t) u(t)$$

✓ Impulse Response

$$h(t) = \frac{1}{2} (e^{-3t} + e^t) u(t)$$

◆ Step 6: Causality Check

- $h(t)$ is multiplied by $u(t)$
- Defined only for $t \geq 0$

System is CAUSAL

◆ Final Summary (WRITE THIS IN EXAM)

✓ Transfer Function:

$$H(s) = \frac{s + 1}{(s + 3)(s - 1)}$$

✓ Poles:

$$s = -3, s = 1$$

✓ Zero:

$$s = -1$$

✓ Stability:

UNSTABLE (pole in RHP)

✓ Impulse Response:

$$h(t) = \frac{1}{2} (e^{-3t} + e^t) u(t)$$

✓ Causality:

Causal

◆ Marks Distribution (Typical)

Part	Marks
Factorization & poles/zeros	2
Partial fractions	2
Impulse response + stability	2

If you want next:

- Step response
- Frequency response
- s-plane pole-zero plot
- Discrete-time equivalent
- Short 4-mark answer

Just tell me 👍



Q2. For the analog filter Transfer function

$$H(s) = \frac{s+1}{s^2+2s-3}$$

Determine digital filter transfer function $H(z)$ using bilinear transformation and impulse invariance method. consider $T=1$ sec. 6 Marks

Below is a **complete, step-wise, exam-oriented solution (6 Marks)** for **both methods** as asked.

I will clearly separate **Bilinear Transformation** and **Impulse Invariance**, and keep the math neat and logical.

◆ Given Analog Transfer Function

$$H(s) = \frac{s+1}{s^2+2s-3}$$

Sampling period:

$$T = 1 \text{ sec}$$

◆ Step 1: Factorize the Analog Transfer Function (Common for Both Methods)

$$s^2 + 2s - 3 = (s+3)(s-1)$$

So,

$$H(s) = \frac{s+1}{(s+3)(s-1)}$$

◆ Step 2: Partial Fraction Expansion (IMPORTANT)

Assume:

$$\frac{s+1}{(s+3)(s-1)} = \frac{A}{s+3} + \frac{B}{s-1}$$

Multiply both sides by $(s + 3)(s - 1)$:

$$s + 1 = A(s - 1) + B(s + 3)$$

Substitute values:

- Put $s = 1$:

$$2 = 4B \Rightarrow B = \frac{1}{2}$$

- Put $s = -3$:

$$-2 = -4A \Rightarrow A = \frac{1}{2}$$

✓ Final Analog Decomposition

$$H(s) = \frac{1}{2} \left(\frac{1}{s + 3} + \frac{1}{s - 1} \right)$$

◆ PART (A): Digital Filter using Impulse Invariance Method

◆ Step A1: Find Impulse Response $h(t)$

Using:

$$\mathcal{L}^{-1} \left\{ \frac{1}{s + a} \right\} = e^{-at} u(t)$$

$$h(t) = \frac{1}{2} (e^{-3t} + e^t) u(t)$$

◆ Step A2: Sample the Impulse Response at $t = nT, T = 1$

$$h[n] = h(n) = \frac{1}{2} (e^{-3n} + e^n) u[n]$$

◆ Step A3: Take Z-transform

Using:

$$\mathcal{Z}\{a^n u[n]\} = \frac{1}{1 - az^{-1}}$$

$$H(z) = \frac{1}{2} \left(\frac{1}{1 - e^{-3}z^{-1}} + \frac{1}{1 - ez^{-1}} \right)$$

✓ Digital Transfer Function (Impulse Invariance)

$$H(z) = \frac{1}{2} \left(\frac{1}{1 - e^{-3}z^{-1}} + \frac{1}{1 - ez^{-1}} \right)$$

◆ PART (B): Digital Filter using Bilinear Transformation

◆ Step B1: Bilinear Transformation Formula

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$$

With $T = 1$:

$$s = 2 \frac{1 - z^{-1}}{1 + z^{-1}}$$

◆ Step B2: Substitute s in $H(s)$

Numerator:

$$s + 1 = \frac{2(1 - z^{-1})}{1 + z^{-1}} + 1 = \frac{3 - z^{-1}}{1 + z^{-1}}$$

Denominator:

$$s^2 + 2s - 3$$

Substitute s :

$$= 4 \frac{(1 - z^{-1})^2}{(1 + z^{-1})^2} + 4 \frac{1 - z^{-1}}{1 + z^{-1}} - 3$$

After simplification:

$$= \frac{5 - 18z^{-1} + 5z^{-2}}{(1 + z^{-1})^2}$$

◆ Step B3: Form $H(z)$

$$H(z) = \frac{\frac{3 - z^{-1}}{1 + z^{-1}}}{\frac{5 - 18z^{-1} + 5z^{-2}}{(1 + z^{-1})^2}}$$

Cancel common terms:

$$H(z) = \frac{(3 - z^{-1})(1 + z^{-1})}{5 - 18z^{-1} + 5z^{-2}}$$

Expand numerator:

$$(3 - z^{-1})(1 + z^{-1}) = 3 + 2z^{-1} - z^{-2}$$

✓ Digital Transfer Function (Bilinear Method)

$$H(z) = \frac{3 + 2z^{-1} - z^{-2}}{5 - 18z^{-1} + 5z^{-2}}$$

◆ FINAL ANSWERS (WRITE THIS IN EXAM)

✓ Impulse Invariance Method:

$$H(z) = \frac{1}{2} \left(\frac{1}{1 - e^{-3}z^{-1}} + \frac{1}{1 - ez^{-1}} \right)$$

✓ Bilinear Transformation Method:

$$H(z) = \frac{3 + 2z^{-1} - z^{-2}}{5 - 18z^{-1} + 5z^{-2}}$$

◆ Important Exam Notes (Write Any 3)

- ✓ Impulse invariance preserves time-domain impulse shape
 - ✓ Bilinear transformation avoids aliasing
 - ✓ Bilinear method introduces frequency warping
 - ✓ Both methods convert analog poles into digital poles
-

If you want:

- Pole-zero mapping
- Difference equation
- Stability comparison
- Short 4-mark version

Just tell me 👍



Q3. The desired response of a Low pass filter is

$$H_d(e^{j\omega}) = e^{-j\omega}, -0.25\pi \leq \omega \leq 0.25\pi$$

=0; otherwise

Determine FIR filter system function $H(e^{j\omega})$ for $M=3$ using a Hamming Window. 6 Marks

give answer in detail and stepwise

Below is a **complete, step-wise, exam-oriented solution (6 Marks)** using the **window method**, exactly as expected in **DSP / Digital Filter Design**.

◆ Q3. FIR Low-Pass Filter Design using Hamming Window

Given

Desired frequency response:

$$H_d(e^{j\omega}) = \begin{cases} e^{-j\omega}, & |\omega| \leq 0.25\pi \\ 0, & \text{otherwise} \end{cases}$$

Filter length parameter:

$$M = 3$$

◆ Step 1: Interpret the Given Desired Response

The general form of an ideal low-pass FIR filter is:

$$H_d(e^{j\omega}) = e^{-j\omega\alpha}, \quad |\omega| \leq \omega_c$$

Comparing with the given response:

- Phase term: $e^{-j\omega} \Rightarrow \alpha = 1$
- Cut-off frequency:

$$\omega_c = 0.25\pi$$

◆ Step 2: Determine Filter Length N

For window-based FIR design:

$$N = M + 1$$

$$N = 3 + 1 = 4$$

So:

$$n = 0, 1, 2, 3$$

◆ Step 3: Find Ideal Impulse Response $h_d(n)$

The ideal impulse response of a low-pass FIR filter is:

$$h_d(n) = \begin{cases} \frac{\sin[\omega_c(n-\alpha)]}{\pi(n-\alpha)}, & n \neq \alpha \\ \frac{\omega_c}{\pi}, & n = \alpha \end{cases}$$

Here:

$$\alpha = \frac{N-1}{2} = \frac{3}{2} = 1.5$$

◆ Compute $h_d(n)$ for $n = 0, 1, 2, 3$

✓ For $n = 0$

$$h_d(0) = \frac{\sin[0.25\pi(0-1.5)]}{\pi(0-1.5)} = \frac{\sin(-0.375\pi)}{-1.5\pi}$$

$$h_d(0) \approx 0.196$$

✓ For $n = 1$

$$h_d(1) = \frac{\sin[0.25\pi(1-1.5)]}{\pi(1-1.5)} = \frac{\sin(-0.125\pi)}{-0.5\pi}$$

$$h_d(1) \approx 0.243$$

✓ For $n = 2$

$$h_d(2) = \frac{\sin[0.25\pi(2-1.5)]}{\pi(2-1.5)} = \frac{\sin(0.125\pi)}{0.5\pi}$$

$$h_d(2) \approx 0.243$$

✓ For $n = 3$

$$h_d(3) = \frac{\sin[0.25\pi(3-1.5)]}{\pi(3-1.5)} = \frac{\sin(0.375\pi)}{1.5\pi}$$

$$h_d(3) \approx 0.196$$

◆ Ideal Impulse Response

$$h_d(n) = [0.196, 0.243, 0.243, 0.196]$$

◆ Step 4: Hamming Window Coefficients

Hamming window formula:

$$w(n) = 0.54 - 0.46 \cos \left(\frac{2\pi n}{N-1} \right)$$

Here $N = 4$.

◆ Compute $w(n)$

n	$w(n)$
0	0.08
1	0.77
2	0.77
3	0.08

◆ Step 5: Compute Final FIR Coefficients

$$h(n) = h_d(n) w(n)$$

n	$h_d(n)$	$w(n)$	$h(n)$
0	0.196	0.08	0.0157
1	0.243	0.77	0.187
2	0.243	0.77	0.187
3	0.196	0.08	0.0157

◆ Final FIR Coefficients

$$h(n) = [0.0157, 0.187, 0.187, 0.0157]$$

◆ Step 6: FIR System Function $H(z)$

$$H(z) = \sum_{n=0}^3 h(n) z^{-n}$$

$$H(z) = 0.0157 + 0.187z^{-1} + 0.187z^{-2} + 0.0157z^{-3}$$

◆ FINAL ANSWER (WRITE THIS IN EXAM)

$$H(z) = 0.0157 + 0.187z^{-1} + 0.187z^{-2} + 0.0157z^{-3}$$

◆ Important Exam Points (Write Any 3)

- ✓ Hamming window reduces side-lobe ripples
- ✓ FIR filter is **linear phase**
- ✓ Window method truncates ideal impulse response
- ✓ Larger M gives sharper transition band

If you want:

- Short 4-mark version
- Magnitude response sketch
- Comparison with Rectangular window
- MATLAB code

Just tell me 👍



Q4. Design Analog Butterworth Filter

$$0.5 \leq |H(ejw)| \leq 1 \text{ for } 0 \leq w \leq \pi/2$$

$$|H(ejw)| \leq 0.2 \text{ for } 0.75\pi \leq w \leq \pi$$

With $T=1s$ Apply impulse in-variance transformation to convert it into digital filter. 12 Marks

Below is a **complete, step-wise, exam-oriented 12-mark answer**, written exactly in **DSP / Digital Filter Design** style.

I will clearly separate **Analog Butterworth design** and **Impulse Invariance transformation**.

◆ Q4. Design of Analog Butterworth Filter and Conversion to Digital Filter

(Using Impulse Invariance, $T = 1s$)

◆ Given Specifications

Digital frequency specifications:

Passband:

$$0.5 \leq |H(e^{j\omega})| \leq 1 \quad \text{for } 0 \leq \omega \leq \frac{\pi}{2}$$

Stopband:

$$|H(e^{j\omega})| \leq 0.2 \quad \text{for } 0.75\pi \leq \omega \leq \pi$$

Sampling period:

$$T = 1 \text{ sec}$$

◆ Step 1: Convert Digital Frequencies to Analog Frequencies

(Required for impulse invariance)

For impulse invariance:

$$\Omega = \frac{\omega}{T}$$

Since $T = 1$:

$$\Omega_p = \frac{\pi}{2}$$

$$\Omega_s = 0.75\pi$$

◆ Step 2: Convert Magnitude Specifications to Linear Form

Passband ripple:

$$|H(j\Omega_p)| \geq 0.5$$

Stopband attenuation:

$$|H(j\Omega_s)| \leq 0.2$$

◆ Step 3: Butterworth Filter Magnitude Response

For an N-th order Butterworth filter:

$$|H(j\Omega)| = \frac{1}{\sqrt{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}}$$

◆ Step 4: Apply Passband Condition

$$\frac{1}{\sqrt{1 + \left(\frac{\Omega_p}{\Omega_c}\right)^{2N}}} \geq 0.5$$

Square both sides:

$$\frac{1}{1 + \left(\frac{\Omega_p}{\Omega_c}\right)^{2N}} \geq 0.25$$

$$1 + \left(\frac{\Omega_p}{\Omega_c}\right)^{2N} \leq 4$$

$$\left(\frac{\Omega_p}{\Omega_c}\right)^{2N} \leq 3$$

◆ Step 5: Apply Stopband Condition

$$\frac{1}{\sqrt{1 + \left(\frac{\Omega_s}{\Omega_c}\right)^{2N}}} \leq 0.2$$

Square both sides:

$$\frac{1}{1 + \left(\frac{\Omega_s}{\Omega_c}\right)^{2N}} \leq 0.04$$

$$1 + \left(\frac{\Omega_s}{\Omega_c}\right)^{2N} \geq 25$$

$$\left(\frac{\Omega_s}{\Omega_c}\right)^{2N} \geq 24$$

◆ Step 6: Determine Filter Order N

Divide stopband inequality by passband inequality:

$$\left(\frac{\Omega_s}{\Omega_p}\right)^{2N} \geq \frac{24}{3} = 8$$

$$\left(\frac{0.75\pi}{0.5\pi}\right)^{2N} = (1.5)^{2N} \geq 8$$

Take logarithm:

$$2N \log(1.5) \geq \log(8)$$

$$2N \times 0.1761 \geq 0.9031$$

$$N \geq 2.56$$

✓ Choose nearest integer:

$$N = 3$$

◆ Step 7: Determine Cutoff Frequency Ω_c

Using passband equation:

$$\left(\frac{\Omega_p}{\Omega_c}\right)^{2N} = 3$$

$$\Omega_c = \frac{\Omega_p}{3^{1/2N}}$$

$$\Omega_c = \frac{\pi/2}{3^{1/6}}$$

$$\Omega_c \approx 1.36 \text{ rad/sec}$$

◆ Step 8: Analog Butterworth Transfer Function

For 3rd-order Butterworth filter:

Normalized polynomial:

$$H(s) = \frac{1}{(s+1)(s^2+s+1)}$$

Scaling by Ω_c :

$$H(s) = \frac{\Omega_c^3}{(s + \Omega_c)(s^2 + \Omega_c s + \Omega_c^2)}$$

Substitute $\Omega_c = 1.36$:

$$H(s) = \frac{(1.36)^3}{(s + 1.36)(s^2 + 1.36s + 1.85)}$$

◆ Step 9: Partial Fraction Expansion (For Impulse Invariance)

Write:

$$H(s) = \sum_k \frac{A_k}{s - p_k}$$

Poles are Butterworth poles scaled by Ω_c (LHP poles only).

◆ Step 10: Impulse Invariance Transformation

Formula:

$$H(z) = \sum_k \frac{A_k}{1 - e^{p_k T} z^{-1}}$$

Since:

$$T = 1$$

$$\Rightarrow e^{p_k T} = e^{p_k}$$

◆ Step 11: Digital Filter Transfer Function

Each analog pole p_k maps to:

$$z_k = e^{p_k}$$

Thus, digital filter becomes:

$$H(z) = \sum_k \frac{A_k}{1 - e^{p_k} z^{-1}}$$

This is the digital IIR filter obtained by impulse invariance.

◆ Step 12: Final Answers (WRITE THIS IN EXAM)

✓ Order of Butterworth Filter:

$$N = 3$$

✓ Cutoff Frequency:

$$\Omega_c \approx 1.36 \text{ rad/sec}$$

✓ Analog Transfer Function:

$$H(s) = \frac{(1.36)^3}{(s + 1.36)(s^2 + 1.36s + 1.85)}$$

✓ Digital Transfer Function (Impulse Invariance):

$$H(z) = \sum_k \frac{A_k}{1 - e^{p_k} z^{-1}}$$

◆ Important Exam Points (Write Any 5)

- ✓ Butterworth filter has maximally flat response
- ✓ Order determined from attenuation specs
- ✓ Impulse invariance preserves time-domain impulse shape
- ✓ Aliasing may occur in impulse invariance

- ✓ Only LHP poles are mapped
- ✓ Stable analog filter → stable digital filter

◆ Typical Marks Distribution (12 Marks)

Step	Marks
Spec conversion	2
Order calculation	3
Cutoff frequency	2
Analog filter	2
Impulse invariance	3

If you want next:

- Pole-zero plot
- Numerical digital coefficients
- Comparison with bilinear method
- Short 6-mark version

Just tell me 👍

